

Rosen — §9.2: n -ary Relations and Their Applications

Selected Exercises

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Exercise 9

The 5-tuples in a 5-ary relation represent these attributes of all people in the United States: name, Social Security number, street address, city, state.

- (a) Determine a primary key for this relation.
- (b) Under what conditions would (name, street address) be a composite key?
- (c) Under what conditions would (name, street address, city) be a composite key?

Exercise 18

How many components are there in the n -tuples in the table obtained by applying the join operator J_3 to two tables with 5-tuples and 8-tuples, respectively?

Exercise 24

Show that if C is a condition that elements of the n -ary relations R and S may satisfy, then

$$\sigma_C(R - S) = \sigma_C(R) - \sigma_C(S).$$

Exercise 25

Show that if R and S are both n -ary relations, then

$$P_{i_1, i_2, \dots, i_m}(R \cup S) = P_{i_1, i_2, \dots, i_m}(R) \cup P_{i_1, i_2, \dots, i_m}(S).$$

Exercise 27

Give an example to show that if R and S are both n -ary relations, then $P_{i_1, i_2, \dots, i_m}(R - S)$ may be different from $P_{i_1, i_2, \dots, i_m}(R) - P_{i_1, i_2, \dots, i_m}(S)$.

Exercise 32

Show that an n -ary relation with a primary key can be thought of as the graph of a function that maps values of the primary key to $(n - 1)$ -tuples formed from values of the other domains.